

Independent Erdos #742

Daily adversarial audit

23 September 2026 | Audited predecessor **320ff91c4b3b1c64e98349788f9f971f65a9ef01**
Independent Murty-Simon proof programme - mathematical and launch-reliability review

Disposition. Material progress survives the audit, but the general theorem is not proved. The raw graph-to-profile bridge now has a standalone derivation and an exact finite regression. Internally, the edge bound is supported through total demand $S \leq 13$ and balanced-complete-bipartite equality through $S \leq 8$. The candidate maximum-degree threshold remains **250/429**; the residual strip and full equality theorem remain open.

What changed in the previous 24 hours

Claim	Audit disposition	Evidence
Raw graph-to-profile bridge	Retained; internally reconstructed	757 D2C graphs; 1,256 roots; 2,722 legal selections; 153 positive-demand selections
Maximum-degree threshold	Unchanged candidate: 250/429	No new lower threshold; prior rational/scalar replay remains current
Zero-demand equality	Retained at internal candidate scope	Both parities; odd $K(k, k+1)$ control preserved
Bounded edge result	Retained through $S \leq 13$	$r=11$ cross-file replay passes after evidence repair
Equality characterization	Retained through $S \leq 8$ only	2,103 $r=8$ equality source populations; zero supplement survivors
$r=12$ support ten	OPEN	42/6,386 source-feasible kernels; supplement solve unfinished

Red-team findings

- 1. Evidence corruption found and repaired.** Two committed files contained truncation-warning text and ellipsis markers: `R8_EQUALITY_CORE_SCREEN.json` and `r11_support9_masks.txt`. The advertised replay initially crashed. The corrupt originals are preserved with `.corrupted-at-320ff91` names; clean outputs were regenerated from `screen_r8_equality.py` and `r11_support9_orbits.cpp`. `verify_r8_pipeline.py` and `audit_r11_pipeline.py` then passed.
- 2. Residual-free helper asymmetry tested.** The earlier $r=8$ equality checker permits residual-free supplement helpers, while later $r=10/r=11$ scripts omit them. An exact helper-aware replay reopened none of the four $r=10$ kernels. An optimistic helper-aware MILP rejected 24/27 $r=11$ kernels; the other three have only the all-zero free type, which discharges no obligation, so their original exhaustive DFS is unaffected. No stored $r=11$ witness population reopened.
- 3. Independence retained.** Search and dependency review found no invocation of the external Erdos742/Erdos742 universal $e+\text{disj}+X$ inequality in the new raw bridge, equality or bounded-demand chain.
- 4. Realizability remains distinct.** The finite profiles are necessary-condition systems. Infeasible rows can exclude graphs only through the audited bridge; surviving abstract rows are not D2C constructions. The $r=11$ cross-file audit does not independently reimplement orbit quotienting or source/supplement algorithms.

Mathematical audit detail

Raw criticality and bridge

The standalone note quantifies over every maximum-degree root and every legal quasi-edge selection. It derives the exact ledger $e(F)=r+t$, source-demand injection, supplement forcing, threshold capacity and radical profile projection. The $a=0$ star case is separated correctly, repairing the earlier strict $0<0$ display without changing either degree assembly.

Fresh execution of `check_raw_profile.py` passed with source SHA-256 `138ef5194f134e1902685692285af1f816a18792dbd337439634bfdc3dfbf2d4`. Coverage includes 8,374 independent BFS edge-deletion checks and both selected/residual supplement branches. Limit: no positive-surplus actual graph was found and only two high-load threshold instances were exercised.

Equality geometry

The $S=0$ proof forces $t=r=e(F)=0$ at equality. The even case yields $K(k,k)$; the odd maximum-degree and sink argument yields $K(k,k+1)$, including the small endpoint. Repeated Boolean codes are not identified with vertices. Balanced complete bipartite graphs remain positive controls rather than excluded cases.

Finite demand closure

For a strict counterexample $S>r+2$. Exact strict closure through $r=11$ therefore supports the edge bound through $S\leq 13$. The repaired high-support replay contains 2,496, 3,273 and 280 strict rows; source survivors 16, 8 and 3; supplement survivors zero. Equality closure through $r=8$ supports equality through $S\leq 8$. All are internal computer-assisted statements.

Remaining mathematical holes

The 42 source-feasible $r=12$ support-ten $(2,2,1^8)$ kernels remain unsolved at the supplement stage. There is no $S\leq 14$ theorem, no threshold below $250/429$, no closure of $n/2<\Delta<250n/429$ and no general equality theorem. The candidate $250/429$ implication is not externally established.

Confidence changes

Item	Before audit	After audit
Raw bridge	Principal un-reconstructed risk	Internally supported; external review still open
$S\leq 13$ finite closure	Claimed internal exact result	Retained after repairing corrupt evidence and helper audit
$S\leq 8$ equality	Claimed internal exact result	Retained after clean regeneration and replay
$r=12 / S\leq 14$	Incomplete	Still incomplete; no confidence promotion
Scheduler recovery	Configuration applied	Still unverified until observed launches

Session launch and utilisation audit

Each trigger is an accounting identity. RAN means evidenced substantive scheduled research; it does not mean the historical 50-minute target was met. Missing time is never inferred from commits, prose or unit count. The 09:14 manual recovery is reported separately.

Trigger	Earliest start	Verified forward	Units	Classification
01:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING
02:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING
03:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING
04:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING
05:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING
06:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING
07:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING
08:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING
09:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING
10:00	10:02:42	35m40s	11	SHORT
11:00	11:01:19	33m21s	20	SHORT
12:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING
13:00	13:58:57	0m00s	0	MISSED/NO FORWARD
14:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING
15:00	15:04:10	32m39s	9	LATE / TARGET N/A
16:00	16:03:51	28m36s	11	LATE / TARGET N/A
17:00	17:00:58	26m37s	13	SHORT
18:00	18:04:09	17m33s	16	LATE / TARGET N/A
19:00	19:58:31	0m00s	0	MISSED/NO FORWARD
20:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING
21:00	21:02:43	28m28s	11	SHORT
22:00	22:00:33	31m02s	9	SHORT
23:00	UNVERIFIED	UNVERIFIED	-	UNVERIFIED/MISSING

Aggregate. 23 scheduled triggers; 23 canonical records after reconciliation (9 existed before audit); 9 fully evidenced deliveries; 8 scheduled RAN; 0 met the historical 50-minute target; 5 short; 3 late-start/target-N/A; 2 zero-research deliveries; 13 missing/unverified. The eight evidenced research windows contain 408m39s of available time and 233m56s of closed forward intervals: 57.25% utilisation over those windows only. Manual recovery adds 11m10s and three units, giving a research lower bound of 245m06s when reported separately. Scheduled units: 100.

Separate activity. One midnight audit, one manual-recovery research run and three administration/configuration episodes. None receives scheduled forward-session credit. No previous preservation segment is evidenced as displacing a later trigger; absent or late launch evidence is the dominant failure.

Prospective cutover. The 40-minute standalone policy starts at 01:00:38 on 23 September. Its first three launches were not due during this audit. The next audit must check each for durable STARTED by trigger+5 minutes and first substantive checkpoint within ten minutes of entry. Configuration and last-run metadata are not recovery evidence.

Focused-session gate and next 24 hours

Gate count: 1/24. Eight useful scheduled sessions ran, but none met the historical 50-minute qualification. Short useful work is not relabelled as no work. The formal go/no-go gate is not due. Its success condition is already provisionally met by the 250/429 threshold improvement, and the bounded-demand system is a credible finite-dimensional obstruction. Recommendation: continue, subject to the stop rules below.

1. Close the 42 $r=12$ kernels

Run the exact helper-aware supplement screen for support-ten $(2,2,1^8)$. Stop with either a complete infeasibility certificate or an explicit survivor list.

2. Independent finite replay

Reimplement one load-bearing $r=11$ orbit/source stage, preferably support nine. Any count mismatch freezes $S \leq 13$.

3. Graph-level stress

Seek actual positive-demand/high-load D2C fixtures and replay every bridge quantity. Bounded nonappearance proves no nonrealizability.

4. Return to the strip

Only after the finite and graph gates, combine the bridge with the high-codegree cluster or prove near-zero-demand stability preserving all balanced complete bipartite controls.

5. Measure launches, not settings

Enumerate every due trigger. Keep RAN, 40-minute compliance and focused-session credit separate; inspect the first three standalone launches individually.

Explicit stop/pivot criteria

Freeze a finite conclusion on any independent orbit/source count mismatch. Freeze 250/429 on any raw bridge counterexample. Reject any equality lemma excluding odd balanced complete bipartite graphs. Do not turn an interrupted solve into evidence. At 24 genuinely credited focused sessions, make the continue/pivot decision from achieved mathematics rather than sunk cost.

Deferred, not performed

This audit did not independently reimplement every orbit canonicalizer or finish the $r=12$ supplement solve. It did not rerun the unchanged 250/429 scalar package because yesterday's exact replay remains current and no threshold change occurred. These limits are part of the evidentiary status, not omissions hidden by an "audit complete" label.

Key files and commits

Purpose	Identifier
Prior audit / threshold	9428d8359b85ab0330906beb3c1e583023ce0aba
Telemetry schema hardening	7499d32d03eee71d985ca6faf30bb590563bca1d
Raw bridge derivation	6d2bb26d604602cbd91c0a57dcdf369aace9b727
Actual-graph bridge regression	a5e1efe9657db4a251df60b97299bb4d25cc79fe
Zero-demand equality	fc05fdbf12a1adac2d57d063232f255300333342
$r=11$ closure	d10d7f7c8c507545fee996ab3c1e0a700314b07b
Audited predecessor	320ff91c4b3b1c64e98349788f9f971f65a9ef01
Audit note	project/research/general_n/2026-09-23-daily-red-team-audit-v1/DAILY_RED_TEAM_AUDIT.md